Named Entity Recognition: A Survey for Indian Languages

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***Abstract*—Named Entity Recognition (NER) is a tool based on principles of Artificial Intelligence (AI) and Natural Language Processing (NLP) for automatically tagging Named Entities from unstructured text. Named Entities, which are generally proper nouns, can be the name of a person, organization, location etc. Named Entity Recognition is a very crucial tool of Natural Language Processing. Some of the application areas of NER include- Information Extraction and Retrieval, Machine Translation, Text Summarization etc. This paper analyses different approaches used in the NER of Indian languages, with emphasis on Hindi. The study compares the different approaches for NER viz. Machine Learning (ML), Rule-based and Hybrid. This study is orchestrated to provide gap analysis in available NER systems for Indian Languages, especially Hindi. It is because existing NER systems are evaluated for accuracies based on predefined datasets, which are not providing universal results on any dataset. Also, there is no standard dataset available for fair comparison of accuracies of Indian Languages. Furthermore, we concentrated our scope to Hindi as it is the official language of India and representative to other Indo-Aryan languages having similar structure, making it a generalized solution. Moreover, the scope of Named Entity tags are limited. Whereas, there is a scope of further classification of Named Entities. As future work, we would like to develop an efficient system, in terms of accuracy, and catering to more Named Entity tags than available presently in Hindi NER tools.**

Keywords—Part of speech, Named Entity Recognition, Machine Learning, Conditional Random Fields, Support Vector Machine, Hidden Markov Models, Hybrid, Rule Based

I. INTRODUCTION

Named Entity Recognition (NER) is a Natural Language Processing (NLP) based tool comprising the principle of Artificial Intelligence, Machine Learning etc, which is used to annotate automatically Named Entities such as Person, Location, Organization, Time, Quantity etc. from given unstructured text. NER has been an essential component in various applications such as Machine Translation, Question Answering, Text Summarization, automatic indexing of books, Efficient Search Engines etc. There are different ways in which the process of NER can be performed. NER task of Hindi language is especially difficult because of its various complexities such as lack of capitalization which is the most important feature of European languages in recognizing as well as classifying named entities. Other challenges include but are not limited to:

1. Hindi is a free word language - they lack a rigid Subject-Verb-Object pattern.
2. Lack of Capitalization makes it difficult for Parts-Of-Speech (POS) taggers as well as NER taggers to identify Named Entities.
3. Word Sense Disambiguation (WSD) Problem of Named Entity: Some named entities can be represented in different ways in different contexts, named entities can be used as common nouns and vice versa. Eg. चंदन can be a name of a person and Sandalwood is also called चंदन in Hindi.
4. For English especially, tools required for pre-processing the text like POS tagger can be used to tag Named Entities, but a study conducted by us has shown that such tools cannot identify all the named entities in Indian Languages.
5. No standard list containing the named entities.

The process of NER starts with unstructured text as input. This text is then pre-processed, eg. removing non-Hindi words from the text (for NER in Hindi). NER systems for the English language are highly efficient, but, in order to accomplish the task for the Hindi language, indigenous tools such as POS tagger, NER tagger are required. Parts of Speech (POS) becomes an important feature while performing the task of NER. POS helps in identifying the context of a word and proves to be a critical feature in the training of Machine Learning models for NER. Standard tools such as Stanford NER are available for European languages. But for the Hindi language, no standard NER tool has yet been developed to the best of our knowledge.

Named Entity Recognition can be implemented predominantly using Machine Learning (ML), Rule-Based NER and Hybrid NER. ML usually makes a model for NER

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using annotated text. Some popular examples of this are Hidden Markov Model (HMM), Conditional Random Fields (CRF), Support Vector Machine (SVM) etc. Tools such as Stanford NER and Spacy use Machine Learning. Rule-Based NER typically uses hand-crafted rules put together by linguists. Hybrid NER uses Rule-Based and ML approaches together.

The output of any NER system is of two types viz. the identification of Named Entity and its categorisation such as Name of Person, Location, Organization etc. There are cases, i.e., an entity is recognised as a Named Entity but the category is not known. In order to perform Information Extraction, Information Retrieval, Machine Translation etc in Hindi and analyses of other Indian languages as well, we need the entities to be identified. In this paper, predominantly we have surveyed existing Indo-Aryan family of languages NER viz., Hindi, Bengali, Marathi and Dravidian languages such as Tamil, Malayalam. Out of this survey we have chosen Hindi as our subject of analysis to further develop an efficient and accurate NER tool in Hindi as it is the official language of India and it is structurally similar to other Indo-Aryan languages as mentioned above. We have also identified the gaps in the existing systems that, if filled, can improve the accuracy in recognition of named entities further.

II. Literature Survey and Related Work

The above mentioned Indo-Aryan and Dravidian languages for our survey have been envisaged to the culmination of approaches of NER that can be categorised broadly into three groups, i.e. Rule-Based, ML and Hybrid approaches. We also observed that Indo- Aryan languages such as Hindi, Bengali, Marathi etc don't need a separate rule-based system when working with ML algorithms (although a higher accuracy is achieved using hybrid approach). However, Dravidian languages being agglutinative in nature require rules (e.g. Morphological Analyser) as well and mostly require a hybrid approach. The following are the approaches surveyed leading to the gap analysis:

A) Rule- Based approach

Association Rule Mining (ARM):

Association rule mining aims to find rules, given a set of transactions. It helps discover relations between variables of a large database. Jain Yadav and Tayal [1] have a proposed a system for Hindi where for every pair of itemsets A and B they find a set of rules where Support >= min (Support) and Confidence >= min (Confidence). Support and Confidence are calculated as follows

$$Support=\frac{Transactions\ having\ items\ A\ and\ B}{Total\ number\ of\ transactions\ in\ the\ database} \quad (1)$$

$$\text{Confidence}=\frac{Transactions\ having\ items\ A\ and\ B\ together}{Transactions\ having\ item\ A\ in\ the\ database} \quad (II)$$

They have identified 3 types of rules- Type 1: Dictionary rule, Type 2: Bi-gram rule and Type 3: feature rule. From the result, it is observed that rules Type 2 and Type 1 combined gives the highest precision of 91.1% and recall of 67.89%.

B) Machine Learning approach

Hidden Markov Model (HMM):

HMM is a statistical Markov model. HMM is a statistical approach to NER where probability of each event depends on the state of the previous event. Chopra Joshi and Mathur [2] have used HMM for NER in Hindi. As they have explained, HMM helps develop language-Independent NER systems. It is also easy to scale and analyze these systems. An F-Measure of 97.14% was obtained on training data of 2343 tokens and testing on 105 tokens. But HMM has the label bias problem. Also, the size of training and testing corpus used in [2] is very limited.

While surveying for NER on the Marathi Language, we found that A. S. Patil, B. V. Pawar and N. V. Patil [3] address the problem of assigning correct named entity class tag to each word using the Hidden Markov Model (HMM), a probabilistic one, trained on a manually tagged corpus for the language. Proposed system in [3] reports an overall F1-Score of 62.70% when no preprocessing was applied whereas it reports an overall F1-Score of 77.79% when preprocessing was applied on the same data. The system described[3] has recognized- the person, locations, numbers and measures well but other named entities are not recognized satisfactorily. [3] uses pre-processing techniques like lemmatization to improve the efficiency which can be a bit costly.

HMM has also been used for Gujarati language by Vora, Vasant and Adhvaryu [4]. The paper mentions no details on the accuracy of their system. Named Entity Recognition in Gujarati is however, rare and very less explored. HMM is easy to implement on any language thus making it desirable for Vora et al. to use it for their NER in Gujarati language. As input they are taking printed versions of Gujarati Text. The drawback they pointed out is that their system wouldn't recognise handwritten documents as writing differs from person to person.

Support Vector Machine (SVM):

Lakshmi G et al. [5] achieved a higher accuracy with Fuzzy SVM over SVM and achieved an F-Score of 88.13. While SVM gave an F-Score of 83.82. SVM is a supervised machine learning algorithm for binary classification. Thus each sentence is given a binary value of either 1 or 0 and a set of feature vectors is created for training data. The integration of Fuzzy module makes their system capable of any multi-class classification.

Conditional Random Fields (CRF):

CRF is another statistical modelling method which, just like HMM, falls into sequence modelling category. CRFs are conditionally trained and can thus easily incorporate a large number of arbitrary and dependent features. This is exactly what Sharma and Goyal [6] have done. They consider a total

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of 29 features such as Context word, Word prefix, Word Suffix, POS information, Gazetteer Lists such as the list of person names, location names, organization names etc. Overall the precision, recall and F-Score they get are 72.78%, 65.82% and 70.45% respectively.

Furthermore, CRF is free from label bias problem, which is not true for HMM. Thus CRF is capable of producing multiple variables that are mutually dependent. Ajees and Idicula [7] take advantage of this in their paper "A Named Entity Recognition System for Malayalam Using Conditional Random Fields". They use tags such as B-PER, I-PER, B-LOC and so on. The accuracy their model achieved is 92.3%.

By combining the outputs of classifiers like Maximum Entropy (ME), Conditional Random Field (CRF) and Support Vector Machine (SVM) using a majority voting approach, Named Entity Recognition (NER) for the Bengali Language is done by Ekbal and Bandopadhyay [8]. They use four major NE tags: Person name, Location name, Organization name and Miscellaneous name tags. The training set consists of about 150K wordforms. Evaluation results of the voted system for the gold standard test set of 30K wordforms have demonstrated the overall recall, precision, and F-Score values of 87.11%, 83.61%, and 85.32%, respectively.

C) Hybrid approach

Sinha focuses on disambiguating Ambiguous Proper Names (APN) in Hindi [9] using CRF. An APN is a name which can also be used as a common noun. As mentioned in section 1, Hindi has a lot of names which are ambiguous in this way. In his paper, Sinha also talks about a need for derivation of a relevant corpus. Subsequently, a relevant corpus has been created tagging it into 3 categories namely: Names (NEP), Words in the dictionary not used as names (NNE) and other words (OTH). A combined output of CRF and Rule-Based where the output is essentially OR-ed gives an overall F-Score of 71.16%.

Furthermore, Das and Dhar [10] not only classify Named Entities like Organization, Person, Location but entities have Before, Internal and End of multiword tags as well that are represented by B-tag, I-tag, E-tag. The aim is to improve the performance of Question Answering, Auto Summarization, Information Retrieval etc with a combination of POS Tagging and Entity Recognition. The Entity Recognition task was done using Conditional Random field (CRF). For training of CRF, they use the ICON 2013 training datasets in open source using the following web resource: CRF3 (https://crfsharp.codeplex.com/sourceControlilatest) classifier which is written in C#. The entity set proposed is Name, Location, Organization, Symbol, Number, Date and Abbreviation. They have a maximum of 0.9156 and a minimum of 0.5714 precision values for symbol and abbreviation entity class respectively and maximum 0.8172 and minimum 0.4444 recall values for these two entity class on the 300 sentences randomly chosen from the ICON 2013 training dataset.

For the Tamil Language, Srinivasagan, Jeyashenbagavalli and Suganthi [11] employ a Hybrid approach that uses both Rule-Based and Hidden Markov Model in succession, which identifies the person, location and organization names. The system [11] identifies unknown entities using statistical Hidden Markov Model. This Named Entity Recognizer shows that the Hybrid approach is better than the statistical model of HMM with overall Precision, Recall and F-Score values as 93.18%, 86.54% and 89.7% respectively through only a limited sized corpus.

A hybrid approach has been used by Patawar and Potey [12] for NER in Marathi language tweets. Twitter being a widely used social media platform naturally is been used in many different languages. In their system, Patawar et al. have used a hybrid of CRF and K-Nearest Neighbour. Initially, the normalized tweets are assigned a confidence value '*cf* ' using a K - value of 4. Then CRF labeler is used to assign a label. CRF calculates the probability, however if 'cf' exceeds then base label is assigned directly. The assigned token is added to the clusters and system uses it for further training. Otherwise, CRF label is assigned to it. Only Location and Names are identified by them with a precision of 39.80 and recall of 85.11 for location and a precision of 59.72 and recall of 25.28 for name tags. CRF makes it easier to add prefixes and suffixes which is necessary for NER in Indian languages.

A Deep Neural based Name Entity Recognizer and Classifier for English Language has been developed by S.P. Singh et al [13]. The system developed by them was tested on 500 sentences and the accuracy varied from 60-70%. The system uses Deep Neural Network for Parts-of-Speech(POS) tagging. POS is an essential clue in identifying the Named Entities in Indian languages. But this approach has not yet been used for Indian Languages. For prediction of Named Entities, first the sentences are chunked into Noun Phrase and Verb Phrase. They have applied handcrafted rules to determine the chunk type of a word (wi) in a Sentence, and then to refer to a memory based classifier in order to check whether the rules satisfy or not. A database (DB2) is maintained which contains chunk type and their corresponding category. The handcrafted rules used also check whether the Entity in question has the first letter in Uppercase or not. This advantage is void in Indian Languages.

III. Stanford NER Tool

Stanford NER tool is a widely recognized tool for Named Entity recognition in various languages such as English, German, Spanish, Chinese etc. For Indian Languages such as Hindi, there is no standard Jar file available as for the languages mentioned above. However Stanford has a provision so that the user can train their own model. To do this, the properties file needs to be edited. We trained our

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model on about 100,000 tokens. The tags identified are NEL, NEP, NETI, NEN, NED. Upon testing with 20 medium sized documents (5400 tokens). The output obtained is as follows:

दिल्ली,NEL प्रदेश,0 कांग्रेस,0 अध्यक्ष,NED अरविंदर,NEP सिंह,0 लवली,0 ने,0 कहा,0 कि,0 सरकार,0 अपने,0 पहले,0 ही,0 बजट,0 में,0 अच्छे,0 दिन,0 लाने,0 का,0 वादा,0 करने,0 के,0 अपने,0 वचन,0 से,0 पीछे,0 हट,0 गई,0

Here the Entities not recognized are marked with a '0'. The precision and recall obtained were 0.45 and 0.5 respectively. The F-score is thus 0.47. This shows that even a standard tool such as Stanford gives only a sub-optimal performance. Hence we have presented a brief of all the gaps identified by us in this paper in the section below.

IV. Gap Analysis

After assessing previous research papers on Named Entity Recognition of Hindi and other Indian languages and testing ourselves the Stanford NER tool for Hindi language, many complications have become apparent. Currently, some of the limitations of existing NER systems are:

1. The accuracy of the NER systems may vary, they're not constant.
2. Even though Rule-Based systems give comparatively high accuracy, they are not robust, as they're created by linguists for a specific domain.
3. Gazetteer list lookup is slow and it is difficult to maintain a list of named entities.
4. Need for the derivation of a relevant corpus - It is observed that NER systems for languages such as English have come up with high accuracy. These systems have an advantage of a huge amount of annotated corpora, and other language resources. However, for Indian languages, there is a lack of resources (e.g. lack of annotated corpora, coarse and Fine-Grained Parts-Of-Speech tags etc). So, developing an NER system with remarkable evaluation measures for such languages is a difficult task.
5. Limited Tagset is considered by most systems.
6. Indian languages need a system that takes into account the context of a word as a large amount of common nouns are used interchangeably with proper nouns.

V. Conclusion and Future Work

This paper studies the different approaches used in the Named Entity Recognition of Indian languages. It is evident from the study that Machine learning approach is more systematic in regards with predicting unknown Entities. However, the accuracy obtained is not as high as rule based systems. Also, many systems have been developed using Deep Learning, Neural Networks etc, for English [13] but this has yet to be implemented for Indian languages. Hence, there is a need for a system that is accurate, robust and predicts unknown named entities using machine learning.

We are trying to create a generalised system tagging more categories than the ones mentioned above in the survey with a hybrid approach that includes an ontology-based lookup, rules and a Machine learning algorithm.

As future work, we are trying to achieve an efficient NER system for Hindi language at Centre for Development of Advanced Computing (C-DAC), Pune that would work on any unstructured text of reporting style of English, in general domains (like newspaper articles, reports, social media, blogs etc.).

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